

Criterion III.2

Functional Compliance of Integration, Securities Features, Reporting and Audit

GOLDBOD TAS

Criterion III.2 Response

Functional Compliance of Integration, Securities Features, Reporting and Audit

Smart Innovations Ghana Limited | Sylvester Akpah

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III.2

Criterion

Functional Compliance of Integration, Securities Features, Reporting and Audit

Government Agency Integrations

The system implements integration touch-points for all six government agencies mandated by the TOR. Each agency integration is built as an independent adapter — replaceable and testable without affecting other services.

Agency	Integration Detail
Minerals Commission	Licence number validation on actor onboarding. Concession zone assignment. Licence expiry tracking with automated 30-day and 7-day alerts. MCAS API + file-based fallback.
Ghana Revenue Authority	Actor tax compliance scoring integrated into KYC. Aggregated gold volume and value data exported monthly. GRA API adapter with CSV fallback.
Bank of Ghana	ISO 20022-format FX reporting for all export transactions. EXPORT event gated on BoG FX approval reference. BoG API adapter.
Ghana Customs Division	Export lot manifests pushed to GCMS before EXPORT event commit. Customs release number stored per CustodyEvent. Cross-border GPS proximity flagging.
Environmental Protection Authority	Concession environmental compliance status per mining site. Mercury-free extraction method flag feeds EPA reporting.
Financial Intelligence Centre	STR/SAR generation workflow with goAML XML format. FIC watchlist synchronisation every 6 hours. Compliance officer disposition dashboard.

Security Features

Authentication & Authorisation

- JWT-based authentication on every API endpoint — tokens expire and require renewal
- Role-based access control: Goldbod User, Miner, Buyer T1/T2, Aggregator, Smelting, Compliance Officer roles
- Role badge visible on all authenticated sessions — confirmed in live screenshot (Paul Ansah: Goldbod User)
- Separate Technical and Financial evaluation tabs — role-separated document access enforced at API level
- Device key registration for field app: Ed25519 key pair, Android Keystore — every CustodyEvent signed

Data Integrity & Non-Repudiation

- Unique Trace ID per gold movement (TRACE-{timestamp}-{sequence}) — tamper-evident audit reference
- Immutable CustodyEvent records — every transfer recorded with actor, GPS, weight, timestamp; never edited
- Ed25519 device signatures on all field-originated events — actor cannot deny submission
- Blockchain layer (Phase 2): SHA-256 hash of every CustodyEvent anchored on Hyperledger Fabric 2.5 — off-chain tampering detectable by hash recomputation
- IPFS content addressing for evidence documents — CID stored on-chain, document substitution detectable

Reporting Capabilities

Transaction History	7 / 30 / 90-day windows. Total balance, transaction count, average per day. Line chart with toggleable date range. 57 live transactions in current 30-day window.
Licence Registry	Recent Buyer T1/T2, Aggregator, and Self-Aggregator licences — Name, Licence No., Issue Date, Expiry Date. Sortable and searchable.
Traceability Audit Trail	Full custody tree per Trace ID — React Flow graph with all actor nodes, weights, purities, and provenance. Exportable for regulatory submission.
Stakeholder Insight	Aggregator breakdown donut chart. Actor count metrics by type. Regional distribution via Geographical Map module.
Regulatory Exports	LBMA Chain of Custody certificate PDF generation. FATF statistical return format. OECD due diligence summary report.
Compliance Dashboard	Open AML flags, STR status tracking, actor risk scores, lot anomaly alerts. Officer disposition workflow with mandatory written justification on clearance.

Audit Trail — Live Evidence

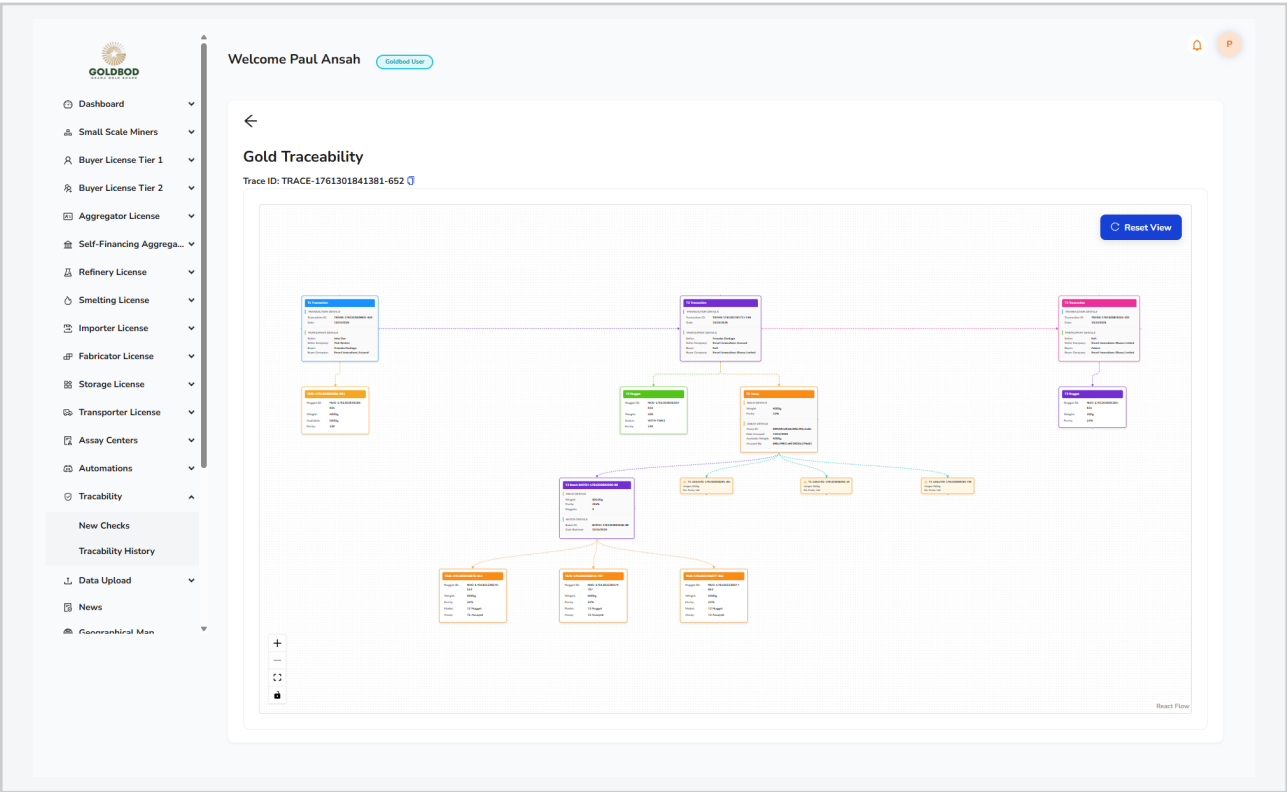


Figure 3 — Full Chain-of-Custody Audit Trail (Trace ID: TRACE-1761301841381-652). Every node in the graph is a permanent, non-editable audit record. Shows source transaction → nuggets → batch consolidation → distributed nuggets with actor identities, weights, purities, and transaction dates at each hop.

EVIDENCE PROVIDED

- ✓ All 6 government agency integrations implemented with API adapters and file-based fallbacks
- ✓ JWT authentication + RBAC — authenticated session confirmed in live screenshots
- ✓ Unique Trace ID system — tamper-evident, copyable, referenceable in STR/SAR reports
- ✓ Immutable CustodyEvent records with Ed25519 device signatures for non-repudiation
- ✓ 30/90-day transaction history reporting with chart and summary statistics
- ✓ Full interactive audit trail via Trace ID — complete custody tree from source to distribution
- ✓ Blockchain hash-anchoring (Phase 2): Hyperledger Fabric 2.5 — tamper detection on all records